

# Jeonggyo Yoon

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## Education

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### **Ph.D. Candidate in Environmental Health Sciences**

Department of Community Environment & Policy, Mel & Enid Zuckerman College of Public Health,  
University of Arizona

Aug. 2021- Present

Advisor: Dr. Chris Lim

Dissertation: "Air pollution, greenness, and cancer-related outcomes"

### **M.S in Humanities and Social Medicine**

Department of Medicine, Graduate School, Gachon University, South Korea

Mar. 2017-Feb. 2019

Advisor: Dr. Yoon-Hyeong Choi

Master's Thesis: "Association between phthalate exposures and walking speed in older Korean adults"

### **B.E in Biomedical Engineering**

Gachon University, College of Health Science, South Korea

Mar. 2012-Feb. 2017

## Research Experience

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### **Graduate Research Associate**

University of Arizona, MEZCOPH, Department of Community, Environment & Policy

Aug. 2021-Present

- **Harmful Algal Blooms and Health:**

- Reviewed and synthesized evidence from 30+ studies on harmful algal blooms and health effects.
- Supported literature review, resource curation, and manuscript revision for a review on HAB aerosols and health (*EBioMedicine* 2023), identifying that most studies relied on proxy exposure metrics rather than direct aerosol measurements and that epidemiological evidence on dose-response relationships and mixture/long-term effects remains limited.

- **Air Pollution Meta-analysis** (manuscript in preparation):

- Screened ~12,000 records, categorized ~200 studies for systematic review and meta-analysis.
- Extracted study-level data and conducted quantitative synthesis of PM2.5 component-specific health effects across four exposure–outcome categories (short-term mortality/morbidity and long-term mortality/morbidity/birth outcomes).

- **Living Systematic Review (LSR) with agentic AI (LLM):**

- Developed an agentic AI pipeline for living systematic reviews, automating literature search, screening, data extraction, risk-of-bias assessment, and meta-analysis. Applied to two active reviews (PM2.5 components × health; greenspace × air pollution) processing over 32,000 records, validated against prior manual reviews.

- **Climate Hazards, Infrastructure, and Equity** (manuscript in preparation):

- Conducted geospatial and equity analyses linking 4.2 million U.S. infrastructure facilities across 84,093 census tracts to 15 climate hazards
- Developed consequence-based classifications for 9 facility types and identified 2,971 triple-risk tracts where hazard exposure, social vulnerability, and low resilience converge.

- Showed that conventional count-based infrastructure screening underweights equity disparities in high-consequence facilities, shifting priority geography when consequence is weighted.
- **Neighborhood Environment and Cancer (Dissertation):**
  - Examined geocoded residential exposures to air pollution and greenspace in relation to obesity-related cancer outcomes in the NIH-AARP Diet and Health Study, a prospective cohort of more than 567,000 participants.
  - Applied time-varying and lagged exposure models with effect modification and mediation analyses, adjusting for demographic and behavioral covariates.
- **Greenspace Morphology and health outcomes:**
  - Investigated associations between greenspace landscape morphology (NLCD 2023) and health outcomes across >60,000 urban census tracts using EPA Level III ecoregion normalization to account for baseline ecological variation. Compared spatial lag and spatial error model specifications and assessed equity implications of greenspace measurement approaches across socioeconomic strata.

*Gachon University College of Medicine, Department of Preventive Medicine*

*Mar. 2020-Aug.2021*

- **HD (Humidifier Disinfectant) and Lung Injury in South Korea**
  - Built a 16-year ecological dataset (176,025 NHIS cases merged with 18-year HD sales records, 9.48M units) to address the absence of population-scale evidence on HD-lung injury association (*IJERPH* 2021).
  - Applied lagged Spearman correlation (0–5 year lags); found temporal concordance between HD consumption and injury incidence ( $r = 0.925$  at 3-year lag), strengthening the causal case for the national HD ban.
- **HD Exposure and Health Conditions Beyond Lung Injury**
  - Analyzed personal exposure assessment data from 4,179 HD claimants, evaluating dose-dependent associations across nine reported health conditions (*BMC Public Health* 2023).

## **Researcher**

*Environmental Health and Epidemiology Laboratory (EHEL), Gachon University*

*Mar. 2019-Feb. 2020*

- **SWEETCH Study (Personalized Risk Communication for EDC Exposure Reduction)**
  - Built full study protocol for a 175-woman, 3-visit cohort: secured dual IRB approval, designed the 25-item EDC exposure questionnaire and codebook, and managed biomonitoring data across multiple EDC biomarkers.
  - Showed risk communication reduced phthalate metabolites short-term, with DEHP metabolites sustained at 5-year follow-up, supporting the efficacy of personalized exposure feedback as an intervention.
- **Investigation of Humidifier Disinfectant (HD) Use in Public Facilities**
  - Conducted on-site investigation in large hospitals and interviewed nurses and administrative assistants to characterize HD exposure conditions, supporting analysis of 162 hospital-exposed HDLI cases from a nationwide dataset ( $n = 6,059$ ).
  - Found 71.2% of previously lung-healthy patients developed new lung disease and 65.2% with existing disease worsened after hospital HD exposure, strengthening evidence that short-term, high-intensity exposure drives acute lung injury.
- **Data Management and Writing for Publication**

- Cleaned and standardized multi-cohort exposure datasets for the HD hospitalization and SWEETCH studies, and produced tables and figures that identified key patterns in the results for publication.
- Co wrote the HD hospitalization paper (*Toxics 2022*) and developed its tables and figures. Built initial datasets, preliminary analyses for the SWEETCH manuscript in preparation.

### **Postgraduate Research Assistant**

*Gachon University College of Medicine, Department of Preventive Medicine*

*Mar. 2017-Feb.2019*

- **Phthalate exposure and frailty (walking speed)**
  - Investigated associations between urinary phthalate metabolites and physical functions (i.e., walking speed, weight loss, frailty) in older Korean adults (*EHP 2023*).
  - Built a repeated-measures dataset (n=1,190, up to 3 visits, 2012-2014) linking five urinary phthalate metabolites to gait speed, and applied GEE-based marginal models, linear mixed-effects models (SAS PROC MIXED), and Bayesian kernel machine regression (BKMR) to test mixture effects.
  - Found MEHHP raised longitudinal odds of slowness by 15% per doubling (OR 1.15, 95% CI: 1.02-1.29), and BKMR mixture modeling showed a 0.047 m/s speed reduction at the 95th vs. 50th percentile of combined exposure

### **Undergraduate Intern**

*Gachon University College of Medicine, Department of Preventive Medicine*

*July 2016 – Feb. 2017*

- **Sarcopenia and Rheumatoid Arthritis/Osteoarthritis: KNHANES**
  - Analyzed DXA-derived skeletal muscle mass and self-reported arthritis data for 11,466 Korean adults aged 40+ using SPSS complex-sample logistic regression, adjusting for 11 covariates (age, sex, income, education, smoking, alcohol, physical activity, muscular training, BMI, hypertension, diabetes).
  - Found a dose-dependent association between sarcopenia severity and rheumatoid arthritis risk (OR 2.14, 95% CI: 1.13-4.05 in sarcopenia vs. high muscle mass, p-trend=0.002), and an elevated osteoarthritis risk in the pre-sarcopenia group (OR 1.45, 95% CI: 1.03-2.05).
  - Conducted the first full research pipeline from data cleaning through manuscript drafting under supervisor guidance, which led to a peer-reviewed publication.

### **Manager (Cycle IV-2) / Field Interviewer (Cycle IV-1/2/3)**

*Korean Society of Environmental Health (KSEH)*

*Nov. 2016 – Mar. 2019*

Korea Environment Industry and Technology Institute (KEITI), “Investigation on Humidifier Disinfectant-associated Lung Injury (Cycle IV)”

- Trained and managed field interviewers affiliated with over 20 local university investigation centers in Korea.
- Coordinated multi-center implementation of a national humidifier disinfectant exposure investigation involving more than 4,500 applicants, including workload distribution, data quality review, and follow-up on incomplete submissions.
- Conducted epidemiologic field investigations on HD exposure cases: Corresponded promptly with all interviewees, coordinated schedule and conducted interviews.

## **Publications & Presentations**

Publications

1. M Hong, MJ Ju, **J Yoon**, W Lee, S Lee, E Jo, S Choi, W Yang, Y Choi. Exposures to humidifier disinfectant and various health conditions in Korean based on personal exposure assessment data of claimants for compensation. BMC Public Health, 23(1), 1800. October 2023.  
<https://doi.org/10.1186/s12889-023-16389-x>
2. CC Lim, **J Yoon**, LB Gerald, AB Ault, S Heo, ML Bell. *Harmful algal bloom aerosols, climate change, and human health: A review*. EBioMedicine. May 2023. <https://doi.org/10.1016/j.ebiom.2023.104604>
3. **J Yoon**, E García -Esquinas, J Kim, JH Kwak, H Kim, S Kim, K Kim, Y Hong, Y Choi. *Urinary phthalate metabolites and slow walking speed in the Korean Elderly Environmental Panel II study*. Environmental Health Perspectives. April 2023. <https://doi.org/10.1289/EHP10549>
4. S Lee, K Han, **J Yoon**, E Jo, W Yang, Y Choi. *New-Onset and Exacerbation of Lung Diseases after Short-Term Exposures to Humidifier Disinfectant during Hospitalization*. Toxics. 2022; 10(7):371.  
<https://doi.org/10.3390/toxics10070371>
5. MJ Ju, **J Yoon**, E Jo, S Lee, J Oh, J Park, W Yang, Y Choi. *A Study on the Regional Distribution of Enrollment Cases Who Reported Humidifier Disinfectant-Related Health Effects*. Journal of Environmental Health Sciences. Aug 2021;47(4):303-9 (in Korean).  
<https://doi.org/10.5668/JEHS.2021.47.4.303>
6. H Ryu, Y Choi, E Kim, J Park, S Lee, **J Yoon**, E Jo, Y Choe, J Heo, W Yang. *Misclassification and characterization of exposure to humidifier disinfectants using a questionnaire*. BMC Public Health. Jul 2021;21(1):1-10. <https://doi.org/10.1186/s12889-021-11459-4>.
7. A Seong, S Heo, **J Yoon**, J Lee, M Choi, J Jeon, J Kim, Kwi Park, J Im, K Ko, J Jung, Y Choi. *Academic Stress and Its Influencing Factors among Medical and Non-Medical Students*. Korean Medical Education Review. Jun 2021, 23(2), 128-138 (in Korean).  
<https://doi.org/10.17496/kmer.2021.23.2.128>.
8. **J Yoon**, M Kang, J Jung, MJ Ju, SH Jeong, W Yang, Y Choi. *Humidifier disinfectant consumption and humidifier disinfectant-associated lung injury in South Korea: A nationwide population-based study*, International journal of environmental research and public health. Jun 2021, 18(11), 6136.  
<https://doi.org/10.3390/ijerph18116136>.
9. K Han, **J Yoon**, E Jo, H Ryu, W Yang, Y Choi, *Case Studies of exposures to humidifier disinfectant in hospitals: Focusing on the exposure assessment of the fourth round of applicants*, Journal of Environmental Health Sciences. Aug 2019, 45(4): 358-369 (in Korean).  
<https://doi.org/10.5668/JEHS.2019.45.4.358>. (**The Best Science and Technology Paper Award**)
10. S Lee, **J Yoon**, J Ock, E Jo, H Ryu, W Yang, Y Choi, *Individual exposure characteristics to humidifier disinfectant according to exposure classification groups –Focusing on 4-1 and 4-2 applicants–*, Journal of Environmental Health Sciences. Aug 2019, 45(4): 370-380 (in Korean).  
<https://doi.org/10.5668/JEHS.2019.45.4.370>
11. H Ryu, E Jo, Y Choi, S Lee, **J Yoon**, JH Kwak, J Park, J Heo, P Kim, W Yang, *Analysis of affecting factors on exposure assessment errors and characteristics of applicants for damage by usage of humidifier disinfectants*, Journal of Environmental Health Sciences. Feb 2019, 45(1): 71-81 (in Korean).  
<https://doi.org/10.5668/JEHS.2019.45.1.71>
12. Y Choi, H Ryu, **J Yoon**, S Lee, JH Kwak, B Han, Y Chu, P Kim, W Yang, *Demographic characteristics and exposure assessment for applicants who have been injured by humidifier disinfectant –Focusing on 4-1 and 4-2 applicants–*, Journal of Environmental Health Sciences. Aug 2018, 44(4): 301-314 (in Korean). <https://doi.org/10.5668/JEHS.2018.44.4.301>

13. **J Yoon**, J Kim, Y Choi, *Associations of sarcopenia with rheumatoid arthritis and osteoarthritis in Korean adults: The Korea National Health and Nutrition Examination Survey 2008-2011*, The Journal of Kinesiology. Jul 2017, 19(3): 61-70 (in Korean). DOI: <https://doi.org/10.15758/jkak.2017.19.3.61>. **(Best Paper Award)**

#### Presentations

1. **J Yoon**, C Lim. *Associations between greenspaces and cancer outcomes in the U.S: the NIH-AARP Diet and Health study*. Poster Presentation at the ISEE (International Society for Environmental Epidemiology) conference, Baltimore, MD (2026.06.03)
2. **J Yoon**, R Jones, J Fisher, G Thurston, C Lim. *Associations between air pollution and cancer survivorship in the U.S: the NIH-AARP Diet and Health study*. Poster Presentation at the ISES (International Society of Exposure Science)/ ISEE (International Society for Environmental Epidemiology) joint conference, Atlanta, GA (2025.08.18)
3. **J Yoon**, C Lim. *Natural hazards and critical infrastructure in the United States*. Oral Presentation at the ISES (International Society of Exposure Science), Montreal, Canada (2024.10.23)
4. **J Yoon**, C Lim, G Thurston. *Associations between Fine Particulate Matter Composition and Sources with Adverse Health Effects: A Systematic Review and Meta-Analysis*. Poster Presentation at the 35th Annual Conference of ISEE (International Society for Environmental Epidemiology), Kaohsiung, Taiwan (2023.09.19)
5. **J Yoon**, C Lim. *Natural hazards and public health infrastructure in the United States*. Poster Presentation at the 35th Annual Conference of ISEE (International Society for Environmental Epidemiology), Kaohsiung, Taiwan (2023.09.19)
6. **J Yoon**, C Lim. *Natural hazards and public health infrastructure in the United States*. Poster Presentation at MEZCOPH Public Health Poster Forum, AZ, US (2023.03.31) **(1<sup>st</sup> place of Research Awards)**
7. **J Yoon**, M Kang, J Jung, MJ Ju, SH Jeong, W Yang, Y Choi. *Humidifier disinfectant sales and humidifier disinfectant-associated lung disease in South Korea: A National Health Insurance Service database*. Poster presentation at Fall Conference of Korean Society of Environmental Health, Online conference, Korea (2020.11.13)
8. **J Yoon**, J Kim, JH Kwak, K Kim, H Kim, S Kim, Y Hong, Y Choi. *Associations between phthalate exposures and walking speed in Korean elderly*. Oral Presentation at Fall Conference of Korean Society of Environmental Health, Seoul, Korea (2018.11.15) **(Best Graduate Student Oral Presentation Award)**
9. **J Yoon**, J Kim, JH Kwak, K Kim, H Kim, S Kim, Y Hong, Y Choi. *Associations between urinary phthalate metabolites and walking speed in elderly adults*. Poster presentation at Spring Conference of Korean Society of Environmental Health and Toxicology, Seoul, Korea, (2018.10.19) **(Best Poster Presentation Award)**
10. **J Yoon**, J Kim, JH Kwak, K Kim, H Kim, S Kim, Y Hong, Y Choi. *Association between phthalate exposure and frailty in Korean older adults*. Poster presentation at Spring Conference of Korean Society of Environmental Health and Toxicology, Seoul, Korea, (2018. 06.01)
11. **J Yoon**, MJ Ju, SG Park, JH Kwak, Y Choi. *Case reports of exposures to humidifier disinfectant in Incheon: Focusing on the exposure assessment of the fourth round of applicants*. Oral Presentation at Spring Conference of Korean Society of Environmental Health, Seoul, Korea (2018.05.31) **(Best Professional Field Interviewer Award)**

12. **J Yoon**, J Kim, Y Choi. *Associations of sarcopenia with rheumatoid arthritis and osteoarthritis in middle-aged and older Korean adults*. Poster presentation at Fall Conference of Korean Society for Preventive Medicine, Busan, Korea (2017. 10. 20) (**Best Poster Presentation Award**)

## Teaching, Mentoring, and Outreach

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### Graduate Teaching Assistant

*University of Arizona, MEZCOPH, Department of Community, Environment & Policy* Aug. 2023-Jan.2024

- Supported instruction for *EHS 575: Environmental and Occupational Health*, an online graduate-level course with 69 students.
- Assisted with course management on D2L, responded to student questions, and provided academic support via Zoom appointments.
- Supported assessment of discussions, quizzes, assignments, and term paper components related to environmental hazard assessment, risk management, and environmental justice.

*University of Arizona, MEZCOPH, Department of Health Promotion Sciences* Aug. 2022-Jan.2023

- Supported instruction for *HPS 300: Public Health in the 21st Century*, an undergraduate course with 56 students.
- Provided student support and feedback on public health writing, discussion activities, assignments, and presentations focused on health disparities, climate change, and other contemporary public health issues.

### Mentoring and Outreach

**Research Mentor**, *STEPS 2 STEM, University of Arizona* June.2025

- Mentored high school students in environmental health research through structured activities including lab tours, paper discussions, data extraction training, and student project development.
- Guided students in interpreting scientific literature on air pollution and health, reviewing study design and results, and preparing a final student exhibit.

**Guest Lecturer**, *Patagonia Union High School, Patagonia, AZ* May.2025

- Presented to 11th- and 12th-grade science students on air pollution and health risks through a community-engaged air quality monitoring context, emphasizing how local data can support environmental health awareness.

## Honors & Awards

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### Scholarships and Travel Support

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|-------------------------------------------------------------------------------------------------|------|
| CRTEC Travel Award, Cancer Research Training and Education Coordination, University of Arizona. | 2025 |
| GPSC Travel Award, Graduate and Professional Student Council, University of Arizona.            | 2025 |
| Dean's Annual Fund Scholarship, College of Public Health, University of Arizona.                | 2023 |
| CRTEC Travel Award, Cancer Research Training and Education Coordination, University of Arizona. | 2023 |

### Research and Presentation Awards

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Best Poster Presentation Award, MEZCOPH Public Health Poster Forum, University of Arizona.                                                                         | 2023 |
| Co-author of award-winning paper, 30 <sup>th</sup> The Best Science and Technology Paper Awards, The Korean Federation of Science and Technology Societies (KOSFT) | 2020 |
| Best Graduate Student Oral Presentation Award, The Korean Society of Environmental Health                                                                          | 2018 |

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| Best Poster Presentation Award, The Korean Society of Environmental Health and Toxicology | 2018 |
| Best Professional Field Interviewer Award, The Korean Society of Environmental Health     | 2018 |
| Best Paper Award, The Journal of Kinesiology                                              | 2018 |
| Best Poster Presentation Award, The Korean Society for Preventive Medicine                | 2017 |

## Professional Memberships

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International Society for Environmental Epidemiology (ISEE)  
 International Society of Exposure Science (ISES)  
 Society for Epidemiologic Research (SER)  
 Korean-American Scientists and Engineers Association (KSEA)

## Skills

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**Statistical Analysis:** Cox proportional hazards, mixed-effects models, logistic regression, BKMR, machine learning, meta-analysis, geospatial analysis, spatial autocorrelation, repeated-measures analysis

**Proficient programs:** SAS, SPSS, R, Stata

**Soft Skills:** Communication, strong work ethic, teamwork, conflict resolution, problem-solving